

The Association Between Procrastination and Insomnia Severity Is Mediated by Binge-Watching

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Abstract

Insomnia is a widespread problem for college students. We examine procrastination as a contributing factor and investigate if binge-watching mediates their association. Despite some limitations, our data showed correlation between insomnia and procrastination with binge-watching serving as a significant mediator, suggesting two potential interventions for insomnia reduction.

Introduction

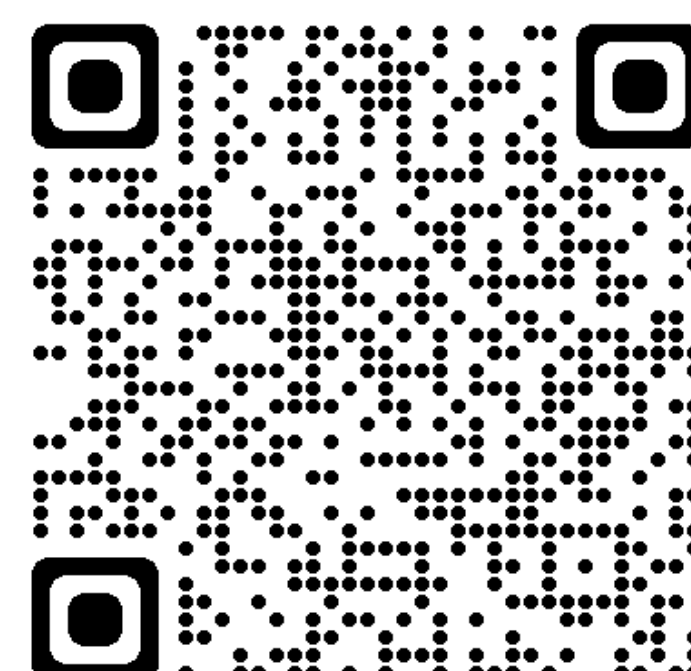
Insomnia is defined by difficulty falling or staying asleep, accompanied with impaired daytime functioning (Sateia, 2014). Relative to the general population, insomnia in college students appears to be particularly problematic, yet relatively understudied (Sivertsen et al., 2019). A potential contributing factor to insomnia is procrastination. Procrastination, defined as the tendency to voluntarily delay the beginning or completion of intended tasks, is prevalent amongst college students, with 30-60% engaging in the behavior (Li et al., 2019; Rabin et al., 2011). Procrastination has been related to shorter sleep duration and poorer sleep quality (Li et al., 2019), but there is limited research on the association between procrastination and insomnia. Beyond establishing a link between the two, it is important to consider possible mediating mechanisms. One potential mechanism is as binge-watching. Procrastination has shown as a significant predictor of binge-watching, defined as watching three or more episodes of television content in one sitting = (Merrill et al., 2019). Research also suggests that binge-watching may contribute to worse sleep quality and insomnia (Exelmans & Van den Bulck, J. 2017). Given these previously observed associations, we hypothesized that binge-watching may serve as a mechanism through which greater procrastination leads to greater insomnia.

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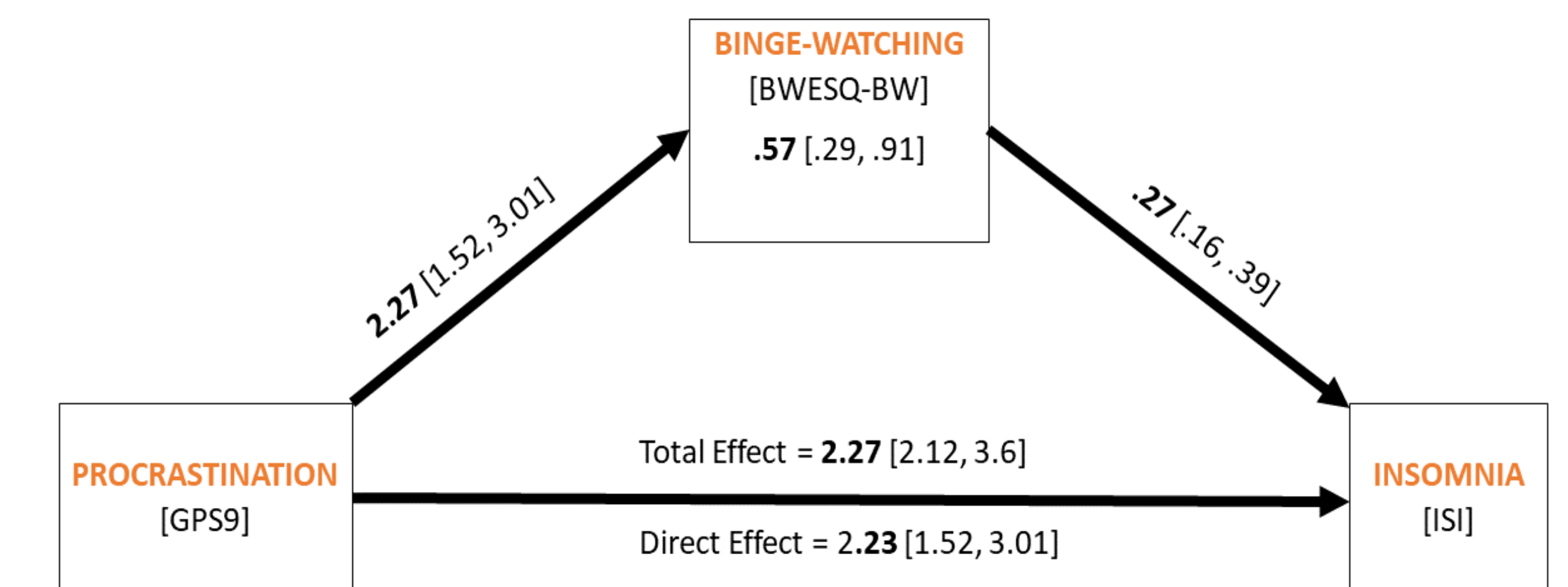


Methods

529 undergraduate students completed an online survey through the Qualtrics platform. Participants were retained if they were between 18 to 25 years of age (N=471). 77.3% were female and 94.5% were Hispanic, with a mean age of 19.65 years old. (SD = 1.68). The survey included the General Procrastination Scale (Sirois, 2019), the “binge-watching” factor from the Binge-Watching Engagement and Symptoms Questionnaire (Flayelle et al., 2020), and the Insomnia Severity Index (Morin et al. 2011).

Results

More than half (56%) of the sampled students met the threshold for insomnia. In support of our hypothesis, procrastination was positively correlated with insomnia severity, ($r(447) = 0.36, p < .01$). Additionally, binge-watching demonstrated a positive correlation with both insomnia severity ($r(447) = 0.31, p < 0.001$) and procrastination ($r(447) = 0.33, p < 0.01$). Procrastination was examined as a predictor of insomnia severity, with binge-watching as a mediator. Binge-watching was found to mediate the procrastination-insomnia relationship (unstandardized indirect effect = 0.57, SE = 0.16, 95% CI [0.29, 0.92]).



Conclusion

Our findings support previous research showing an association between greater procrastination and worse sleep. Based on a mediation analysis, our data demonstrates that binge-watching serves as a potential pathway linking procrastination to greater insomnia. If general efforts to reduce procrastination prove unsuccessful, our results suggest that binge-watching might serve as an intervention. However, we recognize that our study was limited to a cross-sectional analysis and causal pathways cannot be determined. Future research would benefit from longitudinal designs, particularly those involving the effectiveness of interventions. Additionally, although we demonstrated a significant relationship between binge-watching and insomnia severity, we did not explore previously studied factors that might explain this association such as increased pre-sleep cognitive arousal (Exelmans & Van den Bulck, 2017). Furthermore, further research would benefit from using a more ethnically and age diverse sample.

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